

Chat Log

Transformation Code

```
SELECT
  'Diabetes_012' AS variable_name,
  AVG(Diabetes_012) AS mean,
  COUNT(Diabetes_012) AS count,
  COUNT(DISTINCT Diabetes_012) AS unique_count,
  SUM(CASE WHEN Diabetes_012 IS NULL THEN 1 ELSE 0 END) AS null_count,
  MIN(Diabetes_012) AS min_value,
  MAX(Diabetes_012) AS max_value
FROM diabetes_012_health_indicators_brfss2015
UNION ALL
SELECT
  'HighBP' AS variable_name,
  AVG(HighBP) AS mean,
  COUNT(HighBP) AS count,
  COUNT(DISTINCT HighBP) AS unique_count,
  SUM(CASE WHEN HighBP IS NULL THEN 1 ELSE 0 END) AS null_count,
  MIN(HighBP) AS min_value,
  MAX(HighBP) AS max_value
FROM diabetes_012_health_indicators_brfss2015
UNION ALL
SELECT
  'Age' AS variable_name,
  AVG(Age) AS mean,
  COUNT(Age) AS count,
  COUNT(DISTINCT Age) AS unique_count,
  SUM(CASE WHEN Age IS NULL THEN 1 ELSE 0 END) AS null_count,
  MIN(Age) AS min_value,
  MAX(Age) AS max_value
FROM diabetes_012_health_indicators_brfss2015
UNION ALL
SELECT
  'BMI' AS variable_name,
  AVG(BMI) AS mean,
  COUNT(BMI) AS count,
  COUNT(DISTINCT BMI) AS unique_count,
  SUM(CASE WHEN BMI IS NULL THEN 1 ELSE 0 END) AS null_count,
  MIN(BMI) AS min_value,
```



```
MAX(BMI) AS max_value
FROM diabetes_012_health_indicators_brfss2015;
```

Derivation Dialog

user

[CONTEXT]

[TABLE diabetes_012_health_indicators_brfss2015]

Column metadata:

Diabetes_012 (DOUBLE) ---- {'count': 253680, 'unique_count': 3, 'null_count': 0, 'min': 0.0, 'max': 2.0, 'avg': 0.29692131819615264}

HighBP (DOUBLE) ---- {'count': 253680, 'unique_count': 2, 'null_count': 0, 'min': 0.0, 'max': 1.0, 'avg': 0.4290011037527594}

HighChol (DOUBLE) ---- {'count': 253680, 'unique_count': 2, 'null_count': 0, 'min': 0.0, 'max': 1.0, 'avg': 0.4241209397666351}

CholCheck (DOUBLE) ---- {'count': 253680, 'unique_count': 2, 'null_count': 0, 'min': 0.0, 'max': 1.0, 'avg': 0.9626695048880479}

BMI (DOUBLE) ---- {'count': 253680, 'unique_count': 84, 'null_count': 0, 'min': 12.0, 'max': 98.0, 'avg': 28.382363607694735}

Smoker (DOUBLE) ---- {'count': 253680, 'unique_count': 2, 'null_count': 0, 'min': 0.0, 'max': 1.0, 'avg': 0.44316855881425415}

Stroke (DOUBLE) ---- {'count': 253680, 'unique_count': 2, 'null_count': 0, 'min': 0.0, 'max': 1.0, 'avg': 0.04057079785556607}

HeartDiseaseorAttack (DOUBLE) ---- {'count': 253680, 'unique_count': 2, 'null_count': 0, 'min': 0.0, 'max': 1.0, 'avg': 0.09418558814254178}

PhysActivity (DOUBLE) ---- {'count': 253680, 'unique_count': 2, 'null_count': 0, 'min': 0.0, 'max': 1.0, 'avg': 0.7565436770734784}

Fruits (DOUBLE) ---- {'count': 253680, 'unique_count': 2, 'null_count': 0, 'min': 0.0, 'max': 1.0, 'avg': 0.6342557552822453}

Veggies (DOUBLE) ---- {'count': 253680, 'unique_count': 2, 'null_count': 0, 'min': 0.0, 'max': 1.0, 'avg': 0.811419899085462}

HvyAlcoholConsump (DOUBLE) ---- {'count': 253680, 'unique_count': 2, 'null_count': 0, 'min': 0.0, 'max': 1.0, 'avg': 0.05619678334910123}

AnyHealthcare (DOUBLE) ---- {'count': 253680, 'unique_count': 2, 'null_count': 0, 'min': 0.0, 'max': 1.0, 'avg': 0.9510525070955534}



NoDocbcCost (DOUBLE) ---- {'count': 253680, 'unique_count': 2, 'null_count': 0, 'min': 0.0, 'max': 1.0, 'avg': 0.08417691579943236}
GenHlth (DOUBLE) ---- {'count': 253680, 'unique_count': 5, 'null_count': 0, 'min': 1.0, 'max': 5.0, 'avg': 2.5113923052664773}
MentHlth (DOUBLE) ---- {'count': 253680, 'unique_count': 31, 'null_count': 0, 'min': 0.0, 'max': 30.0, 'avg': 3.1847721538946705}
PhysHlth (DOUBLE) ---- {'count': 253680, 'unique_count': 31, 'null_count': 0, 'min': 0.0, 'max': 30.0, 'avg': 4.2420805739514345}
DiffWalk (DOUBLE) ---- {'count': 253680, 'unique_count': 2, 'null_count': 0, 'min': 0.0, 'max': 1.0, 'avg': 0.16822374645222327}
Sex (DOUBLE) ---- {'count': 253680, 'unique_count': 2, 'null_count': 0, 'min': 0.0, 'max': 1.0, 'avg': 0.4403421633554084}
Age (DOUBLE) ---- {'count': 253680, 'unique_count': 13, 'null_count': 0, 'min': 1.0, 'max': 13.0, 'avg': 8.032119205298013}
Education (DOUBLE) ---- {'count': 253680, 'unique_count': 6, 'null_count': 0, 'min': 1.0, 'max': 6.0, 'avg': 5.050433617155472}
Income (DOUBLE) ---- {'count': 253680, 'unique_count': 8, 'null_count': 0, 'min': 1.0, 'max': 8.0, 'avg': 6.053874960580258}

Sample data:

| Diabetes_012 | HighBP | HighChol | CholCheck | BMI | Smoker | Stroke | HeartDiseaseorAttack | PhysActivity | Fruits | Veggies | HvyAlcoholConsump | AnyHealthcare | NoDocbcCost | GenHlth | MentHlth | PhysHlth | DiffWalk | Sex | Age | Education | Income |
0| 0.0 | 1.0 | 1.0 | 1.0 | 40.0 | 1.0 | 0.0 | 0.0 | 0.0 | 0.0 | 1.0 | 0.0 | 1.0 | 0.0 | 5.0 | 18.0 | 15.0 | 1.0 | 0.0 | 9.0 | 4.0 | 3.0 |
1| 0.0 | 0.0 | 0.0 | 0.0 | 25.0 | 1.0 | 0.0 | 0.0 | 1.0 | 0.0 | 0.0 | 0.0 | 0.0 | 1.0 | 3.0 | 0.0 | 0.0 | 0.0 | 0.0 | 7.0 | 6.0 | 1.0 |
2| 0.0 | 1.0 | 1.0 | 1.0 | 28.0 | 0.0 | 0.0 | 0.0 | 0.0 | 1.0 | 0.0 | 0.0 | 1.0 | 1.0 | 5.0 | 30.0 | 30.0 | 1.0 | 0.0 | 9.0 | 4.0 | 8.0 |
3| 0.0 | 1.0 | 0.0 | 1.0 | 27.0 | 0.0 | 0.0 | 0.0 | 1.0 | 1.0 | 1.0 | 0.0 | 1.0 | 0.0 | 2.0 | 0.0 | 0.0 | 0.0 | 0.0 | 11.0 | 3.0 | 6.0 |
|
4| 0.0 | 1.0 | 1.0 | 1.0 | 24.0 | 0.0 | 0.0 | 0.0 | 1.0 | 1.0 | 1.0 | 0.0 | 1.0 | 0.0 | 2.0 | 3.0 | 0.0 | 0.0 | 0.0 | 11.0 | 5.0 | 4.0 |
|

[GOAL]

What are the main characteristics of the dataset?

[OUTPUT]

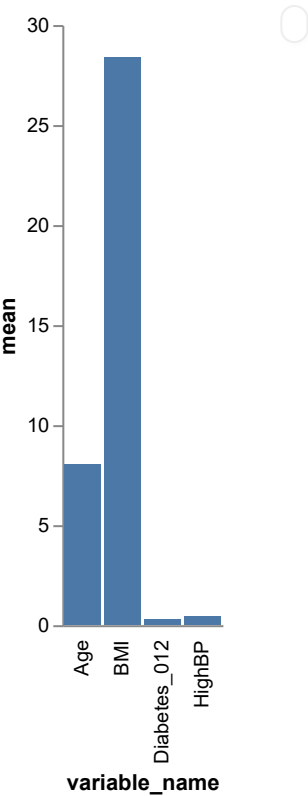
assistant

```
```json
{
 "mode": "overview",
 "recommendation": "To understand the main characteristics of the dataset, we need to provide an overview of the distribution of key variables. This includes understanding the central tendency and dispersion of continuous variables and the frequency distribution of categorical variables.",
 "output_fields": ["variable_name", "mean", "count", "unique_count", "null_count", "min_value", "max_value"],
 "chart_type": "bar",
 "visualization_fields": ["variable_name", "mean"]
}
```

```
```sql
SELECT
  'Diabetes_012' AS variable_name,
  AVG(Diabetes_012) AS mean,
  COUNT(Diabetes_012) AS count,
  COUNT(DISTINCT Diabetes_012) AS unique_count,
  SUM(CASE WHEN Diabetes_012 IS NULL THEN 1 ELSE 0 END) AS null_count,
  MIN(Diabetes_012) AS min_value,
  MAX(Diabetes_012) AS max_value
FROM diabetes_012_health_indicators_brfss2015
UNION ALL
SELECT
  'HighBP' AS variable_name,
  AVG(HighBP) AS mean,
  COUNT(HighBP) AS count,
  COUNT(DISTINCT HighBP) AS unique_count,
  SUM(CASE WHEN HighBP IS NULL THEN 1 ELSE 0 END) AS null_count,
  MIN(HighBP) AS min_value,
  MAX(HighBP) AS max_value
```



```
FROM diabetes_012_health_indicators_brfss2015
UNION ALL
SELECT
  'Age' AS variable_name,
  AVG(Age) AS mean,
  COUNT(Age) AS count,
  COUNT(DISTINCT Age) AS unique_count,
  SUM(CASE WHEN Age IS NULL THEN 1 ELSE 0 END) AS null_count,
  MIN(Age) AS min_value,
  MAX(Age) AS max_value
FROM diabetes_012_health_indicators_brfss2015
UNION ALL
SELECT
  'BMI' AS variable_name,
  AVG(BMI) AS mean,
  COUNT(BMI) AS count,
  COUNT(DISTINCT BMI) AS unique_count,
  SUM(CASE WHEN BMI IS NULL THEN 1 ELSE 0 END) AS null_count,
  MIN(BMI) AS min_value,
  MAX(BMI) AS max_value
FROM diabetes_012_health_indicators_brfss2015;
---
```



AI generated results can be inaccurate, inspect it!



sample data from **view_gsfe** (7 columns × 4 rows)

[show column stats](#)

variable_name	mean	count	unique_count	null_count	min_value	max_value
Diabetes_012	0.2969213182	253680	3	0	0	2
HighBP	0.4290011038	253680	2	0	0	1
Age	8.0321192053	253680	13	0	1	13
BMI	28.3823636077	253680	84	0	12	98